NATURAL LANGUAGE PROCESSING AND COMPUTATIONAL LINGUISTICS

CASE STUDY: COHERE – AN ENTERPRISE NLP STARTUP

NAME: P. Ganesh Prakadesh

REG.NO: 23MIC0020

1. Introduction

Cohere is an artificial intelligence startup that focuses on Natural Language Processing (NLP), Large Language Models (LLMs), generative AI, semantic search, information retrieval, and enterprise AI. The company was started in 2019 in Toronto, Canada. Has become a major global AI company. Other companies often concentrate on consumer-facing chatbots. Coheres' main goal is to offer organizations strong language-processing technologies that can be used in their own apps and business processes. Cohere makes more than the required ₹5 crore every year. Some reports say its yearly income is in the hundreds of millions of dollars.

2. Origin and Founders

Aidan Gomez, Nick Frosst, and Ivan Zhang started Cohere. The history of Cohere matters a lot to the world of NLP because Cohere is tied to how the Transformer architecture was made.

Aidan Gomez worked as a researcher on the 2017 paper called "Attention Is All You Need." That paper is what brought the Transformer architecture to life. The Transformer architecture uses something called self-attention to figure out how words in a sentence relate to each other. This Transformer architecture eventually became the base for famous language models like GPT, BERT, and the models made by Cohere.

After doing research, in machine learning Aidan Gomez and his partners started Cohere. They wanted to make sure advanced language-model technology could actually help businesses. Of just building a chatbot for regular people to use, they saw a way to help companies. They wanted to help organizations read and make sense of piles of internal documents, reports, databases, and other messy information.

3. Founder Details

Aidan Gomez is the co-founder and CEO of Cohere. Aidan Gomez is a computer scientist and machine-learning researcher whose work is strongly linked to the Transformer architecture. Aidan Gomez’s research background gave Cohere a technical base in modern NLP and deep learning.

Nick Frosst is another co-founder who has a background in machine learning and neural networks. Nick Frosst has worked a lot in intelligence research and has helped build Cohere’s technical foundation.

Ivan Zhang, the co-founder, also has a background in machine learning and technology. Together the three founders, Aidan Gomez, Nick Frosst, and Ivan Zhang, combined their research expertise to build AI systems that are useful for businesses.

4. Products and Services

Cohere makes different tools that help with natural language processing. One of the important tools from Cohere is the Command family of large language models. Command models can do things like writing text, making summaries, answering questions, sorting data, thinking through problems, and pulling out specific facts. Cohere built Command models for big companies that need to handle a lot of text at once.

Cohere also made Command R. Command R is a group of models made for things called Retrieval-Augmented Generation (RAG). RAG lets an AI look up facts from a library of knowledge before the AI gives an answer. This is great for companies because the AI can read company documents instead of just knowing what it learned during its initial training.

Another big tool is Embed. Embeds turn text into numbers that are called embeddings. These embeddings help the computer understand the meaning behind words. For instance, "How do I reset my password?" I forgot my login credentials." Use different words, but the meaning is the same. An Embed model can see that these mean the thing. This helps businesses build search tools, suggestion systems, and ways to find documents.

Cohere also has Rerank. Rerank makes search tools work better. When a search tool finds a bunch of documents that might be right, Rerank looks closely at the user's question. Those documents to find the best match. This is very helpful for RAG systems because if the RAG system finds information, the final answer from the AI will also be bad.

Cohere has also moved into the world of business AI with North. North is a platform that brings together AI, company search, and smart workflows. North is not just for chatting with an AI. Instead, Cohere designed North to help workers use information better and finish business tasks automatically.

5. Customers and Target Market

Cohere mainly aims at businesses, governments, and large organizations, not at consumers. Cohere technology can be used by banking, financial services, healthcare, telecommunications, technology, manufacturing, professional services, and government sectors.

The enterprise market fits Cohere well because large organizations hold huge amounts of information. Banks may have rules and internal policies, hospitals may have medical and administrative paperwork, and multinational companies may hold thousands of reports and operational documents.

For example, an employee at an institution might ask, "What are the current requirements for approving an international corporate loan?" Than searching manually through many documents, a Cohere system can understand the question, pull the relevant documents, rank them by relevance, and use a language model to provide an answer.

6. NLP Technologies Used

Coheres products use key NLP and machine-learning tools. The main building block is the Transformer architecture, which relies on self-attention to see how tokens relate to one another. Above this architecture sit language models that can understand and produce natural language.

Embeddings turn the meaning of words into numbers. Semantic search looks for information based on the meaning of just matching keywords. Reranking makes the returned results more relevant. RAG links language models to knowledge sources.

The entire flow looks like this: User Query → Semantic Search → Document Retrieval → Reranking → Relevant Context → Language Model → Generated Answer

This mix lets Coheres technology do more than just simple text generation.

7. Major Innovation

What makes Cohere stand out much is how Cohere focuses on big businesses instead of just regular people. A lot of AI companies spend all their time making chatbots for users. Cohere does things differently by giving companies the tools they need to build their own AI systems.

One big thing Cohere does is use RAG to connect LLMs with private company data. A regular language model might not know about a company's rules or secret files. Cohere uses special retrieval tools so an AI system can look through a company's own files first. This helps the language model get the facts before the language model even starts to write an answer.

This makes the AI more helpful for businesses because the answers use real, up-to-date information from the company. It shows that NLP is changing. It is not about writing text anymore. Now NLP is about finding, reading, and thinking through piles of information.

8. Business Model and Revenue

Cohere is mainly a B2B enterprise AI and API-based company. I see that companies can add Cohere’s language models and NLP tools to their software so staff do not need a separate consumer app. Major investors have put a lot of money into Cohere, giving Cohere a valuation in the multi-billion-dollar range. Revenue has grown a lot because more firms are adopting AI. New data shows Cohere earns over one hundred million dollars each year, more than the five crore minimum required for this study. Cohere’s model proves that NLP can be a business even without a mass-market app. I think businesses are ready to pay for AI that keeps their data safe and boosts productivity.

9. Example of a Real-World Application

Consider a bank that has lots of workers and millions of internal papers. An employee who works in making sure everything is done right might need to check if a certain money move follows the company's rules.

A system that uses Cohere could start by turning the employee's question into a meaning-based version using a model that creates embeddings. It could then look through the company's papers.

Find ones that might be connected. A reranking system could then pick the papers. After that the important parts of those papers could be given to a command model through RAG.

The language model would then create a response in language using the information it found. So many different NLP tools work together. Embeddings understand what things mean; retrieval finds the information, reranking picks the best sources, RAG gives context, and the language model creates the final answer.

10. Conclusion

Cohere is an example of a successful NLP startup because its development is directly connected to one of the most important breakthroughs in modern NLP: the Transformer architecture. Founded in Toronto in 2019 by Aidan Gomez, Nick Frosst, and Ivan Zhang, Cohere has developed enterprise-focused language models and NLP technologies, including Command, Embed, Rerank, and North.

Cohere's main customers are businesses and organizations that need to work with amounts of internal information. Cohere's technology enables them to perform searches, retrieve relevant documents, generate summaries, answer questions, and automate workflows using natural language.

Cohere's major innovation is its focus on enterprise NLP, the combination of LLMs, embeddings, information retrieval, reranking, and RAG. With revenue exceeding ₹5 crore per year and a rapidly expanding enterprise AI business, Cohere demonstrates how fundamental NLP research can be transformed into commercially valuable technology.